Bayesian Discrete-Time Survival Modeling with Sensitivity Analysis for Informative Censoring: An Application to ICU Mortality

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1. Introduction

Accurate mortality risk profiling in the Intensive Care Unit (ICU) is essential for clinical decision-making and resource allocation. Essentially all survival models rely on the non-informative independent censoring assumption, that is, the censoring mechanism (e.g., hospital discharge) carries no information about the underlying survival process beyond observed covariates (Kalbfleisch and Prentice, 2003). In the ICU, this assumption is immediately suspect. Patient discharge is a purposeful clinical decision, exactly dependent on patient health status, creating a dependence between censoring C_i and unobserved failure time T_i, which breaks the standard assumption and biases hazard estimates and covariate effects.

Tsiatis (1975) established a fundamental identifiability result: from the observed data alone (y_i, δ_i), marginal distributions of T_i and C_i cannot be separately identified without structural assumptions on their joint distribution. Diagnostics based on the observed data alone cannot distinguish informative from non-informative censoring, motivating sensitivity analyses.

This paper addresses the central question: *how robust are Bayesian posterior inferences about ICU mortality risk factors to departures from the independent censoring assumption?* We make three contributions within a discrete-time framework using person-period format (Allison, 1982). First, we formalize the independent censoring assumption in the missing data taxonomy, deriving explicit bias formulas for hazard estimation and covariate effects (Section 3.2).[1] Second, we develop a sensitivity analysis following Scharfstein et al. (1999) (and course notes) by introducing γ under a selection model approach that indexes the departure from MAR (Section 3.5). Third, we compare this with a shared frailty model (Section 3.4) and demonstrate through both simulation and application to 57,694 MIMIC-III patients (Section 5) that frailty is weakly identified from data sparsity in panel data. This paper provides a framework for assessing robustness of inferences to this pervasive issue.

1. Full derivations, simulation study details, and supplementary cohort tables are available at `https://github.com/adamkurth/bayes-mimic-survival`.

2. Data Structure & Notation

2.1 MIMIC-III Cohort

Let $i = 1, \ldots, N$ index unique ICU admissions from MIMIC-III. Define $T_i > 0$ as latent time to death, $C_i > 0$ as the censoring time (hospital discharge), and $y_i = \min(T_i, C_i)$ as the observed follow-up time. The failure indicator is $\delta_i = \mathbb{I}(T_i \leq C_i)$, $\delta_i = 1$ when patient died in the ICU, or $\delta_i = 0$ if discharged. $\mathbf{x}_i \in \mathbb{R}^P$ is a vector of static baseline covariates measured at admission. The observed data for patient i is thus $(y_i, \delta_i, \mathbf{x}_i)$. The data covariate characteristics are described in more detail in Section 5, and Table 3.

Discretization. We partition continuous time into discrete intervals of width Δ: The intervals are $[0, \Delta)$, $[\Delta, 2\Delta), \ldots, [(K-1)\Delta, K\Delta)$ with $t = 1, \ldots, K$. The discrete failure time is $\tilde{T}_i = \lceil T_i/\Delta \rceil$ and $\tilde{C}_i = \lceil C_i/\Delta \rceil$. Our primary analysis uses daily discretization ($\Delta = 1$ day), over 30-day window ($K = 30$), motivated by the standard use of 30-day ICU outcome metric, the median follow-up of ≈ 6 days in MIMIC-III, so the window captures the clinically relevant risk period. We drop the tildes hereafter, so T_i and C_i refer to the discretized times. Our simulation study confirms stability across $\Delta \in \{1, 3, 7, 14\}$.

Person-period expansion. Following Allison (1982), the data is reconstructed into a "person-period" format: each patient i contributes one row per interval $t = 1, \ldots, K_i$ where $K_i = \min(T_i, C_i)$ is the last interval for which patient i is at risk. At time $t = K_i$, the binary outcome is $y_{i,K_i} = \mathbb{I}(\delta_i = 1)$, either 1 if patient i dies or 0 if censored. These are absorbing states, so patients exit the risk set thereafter. The risk set at time t is $\mathcal{R}_t = \{i : y_i \geq t\} = \{i : T_i \geq t, C_i \geq t\}$ and the censored set is $\mathcal{C}_t = \{i : C_i < t\}$. The person-period expansion yields $R = 509,944$ rows.

2.2 Covariate Matrix

The design matrix $\mathbf{X} \in \mathbb{R}^{N \times P}$ with $P = 17$ includes baseline covariates measured at ICU admission (Table 3): age (standardized), sex, emergency admission, ICD-9 codes for sepsis, heart/respiratory failure, acute kidney injury, pneumonia, diabetes, chronic liver disease, malignancy, and COPD, Glasgow Coma Scale impairment (GCS < 12), vasopressor usage, and binary indicators for mechanical ventilation, renal replacement therapy, and positive blood cultures. All covariates are binary except standardized age.

Table 3 presents baseline covariates by prevalence across outcomes. Patients who died are substantially older (mean age 70 vs. 61 for discharged), more likely to have been admitted emergently, and have markedly higher rates of sepsis, respiratory failure, and vasopressor use. The discharged-alive group was younger with lower comorbidity burden, consistent with the hypothesis that discharge represents a positive prognostic signal—exactly the pattern that makes informative censoring plausible.

2.3 Censoring Pattern

Figure 1 shows the empirical follow-up distribution and risk set evolution. The follow-up distribution is heavily front-loaded with a mode at 4–6 days and by day 30, only approximately 3,200 patients remain at risk. Furthermore, discretization past $\Delta = 1$ daily intervals, would result in a substantial loss of information. This front-loaded discharge pattern underscores

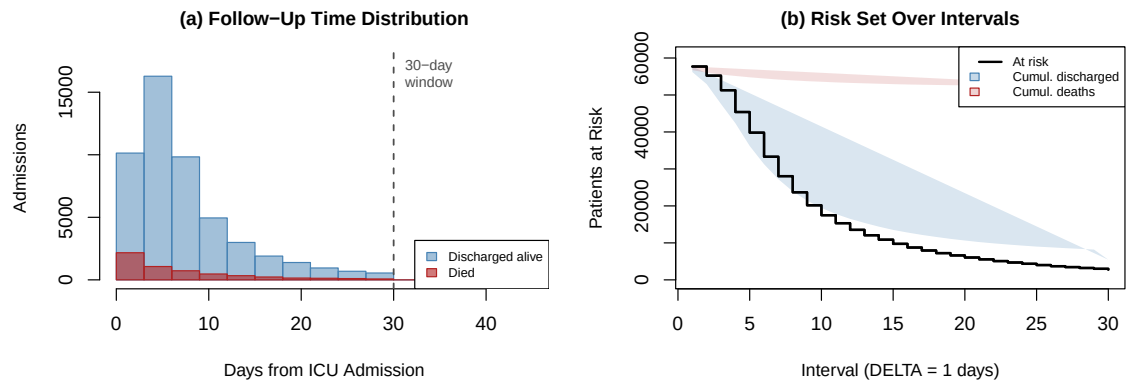


Figure 1: Empirical follow-up distribution and risk set evolution for the MIMIC-III cohort ($N = 57,694$; $\Delta = 1$ day, $K = 30$). **(a)** Histogram stratified by outcome: death (red), discharge (blue). The dashed vertical line marks the 30-day observation window. **(b)** Risk set size (black) with cumulative discharges (blue) and deaths (red). The widening gap illustrates that the risk set depletes primarily through discharge, especially in the first week.

the importance of the sensitivity analysis. If discharged patients would have survived longer under continued observation, the standard analysis overestimates later-interval hazards per the bias formula derived in Section 3.3.

3. Methodology

3.1 Selection Factorization

Censoring is treated as a missing-outcome problem. The failure indicator $\delta_i = \mathbb{I}(T_i \leq C_i)$ already partitions patients into those whose mortality is observed ($\delta_i = 1$) and those censored by discharge ($\delta_i = 0$). The observed data $\mathcal{D}^o = \{(y_i, \delta_i, \mathbf{x}_i)\}_{i=1}^N$ is a coarsened version of the complete data $\mathcal{D}^c = \{\mathcal{D}^o, \mathcal{D}^m\}$, where $\mathcal{D}^m$ contains the unobserved mortality outcomes of censored patients. The joint distribution of the complete data uses the selection factorization (Rubin, 1976; Oganisian, 2026):

$$f(\delta_i, y_i, \mathbf{x}_i \mid \omega) = \underbrace{f(\delta_i \mid y_i, \mathbf{x}_i; \xi)}_{\text{Selection model}} \cdot \underbrace{f(y_i \mid \mathbf{x}_i; \eta)}_{\text{Outcome model}} \cdot \underbrace{f(\mathbf{x}_i; \theta)}_{\text{Covariate model}} \tag{1}$$

where $\omega = (\xi, \eta, \theta)$ are distinct independent a priori parameters governing each component. The selection model $f(\delta_i \mid y_i, \mathbf{x}_i; \xi)$ describes the censoring mechanism, the outcome model is of inferential interest, and the covariate model is ancillary. The censoring mechanism is ignorable for inference about T_i if the selection model is MAR, where $M_i \perp Y_i \mid X_i$ implies $f(\delta_i \mid y_i, \mathbf{x}_i; \xi) = f(\delta_i \mid \mathbf{x}_i; \xi)$. Under MAR, inference proceeds using only observed data. Otherwise, censoring is informative and classified as Missing Not at Random (MNAR): $M_i \not\perp Y_i \mid X_i$, requiring joint modeling. In the ICU, MNAR is the more plausible scenario, as discharge decisions may reflect systematically different mortality risk than those who remain, even conditional on $\mathbf{x}_i$.

3.2 Standard Likelihood & Censoring Assumption

The discrete-time hazard for patient i at time t is $h_{i,t} = P(T_i = t \mid T_i \geq t, \mathbf{x}_i)$, with survival function $S_i(t) = P(T_i > t \mid \mathbf{x}_i) = \prod_{s=1}^{t}(1 - h_{i,s})$. The full data likelihood involves the joint density of (y_i, δ_i) depending both on failure and censoring parameters $\boldsymbol{\theta}, \boldsymbol{\phi}$ respectively (Kalbfleisch and Prentice, 2003, Section 3.2):

$$\mathcal{L}_i^{\text{full}}(\boldsymbol{\theta}, \boldsymbol{\phi}) = \left[f_T(y_i \mid \mathbf{x}_i)\, S_C(y_i \mid \mathbf{x}_i)\right]^{\delta_i} \left[S_T(y_i \mid \mathbf{x}_i)\, f_C(y_i \mid \mathbf{x}_i)\right]^{1-\delta_i} \tag{2}$$

Assumption: *[Independent Right-Censoring; K&P Section 2.4] $T_i \perp C_i \mid \mathbf{x}_i$: knowing the censoring time provides no additional information about survival beyond $\mathbf{x}_i$.*

Under Assumption 3.2, (2) factors so that inference about $\boldsymbol{\theta}$ proceeds without modeling censoring. In person-period form: $\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^{N}\prod_{t=1}^{K_i} h_{i,t}^{y_{i,t}}(1 - h_{i,t})^{1-y_{i,t}}$, where each (i, t) term is an independent Bernoulli trial. This corresponds to MAR in the selection factorization (1). Since time-varying information is likely far more informative for discharge decisions, and $\mathbf{x}_i$ only contains baseline covariates, MAR is unlikely to hold ($T_i \not\perp C_i \mid \mathbf{x}_i$), motivating the methods below.

3.3 Consequences of Informative Censoring

When the independent censoring assumption fails, two consequences follow. First, systematic discharge of healthier patients inflates the observed hazard. Let $\lambda_t = |\mathcal{C}_t|/|\mathcal{R}_t|$ be the proportion of patients censored at t, and $\Delta_t = h_t^* - \hat{h}_t$ the difference between the true marginal hazard h_t^* and the observed hazard $\hat{h}_t$. The bias formula (derived in the supplementary material) shows:

$$\hat{h}_t \approx h_t^* + \frac{\lambda_t \Delta_t}{1 - \lambda_t} \quad \text{where} \quad \hat{h}_t = \frac{\sum_{i \in \mathcal{R}_t} \mathbb{I}(T_i = t)}{|\mathcal{R}_t|} \tag{3}$$

The bias is positive when $\lambda_t, \Delta_t > 0$ (censored patients are healthier), but Δ_t could be negative if censored patients are sicker (e.g., hospice discharge).

Second, covariate effects $\boldsymbol{\beta}$ are confounded by the censoring process. Scharfstein et al. (1999) formalizes this as a failure of the semiparametric estimation equations when MAR is violated, where the score function for $\boldsymbol{\beta}$ is biased proportional to a non-identifiable selection parameter (see supplementary derivation). If the selection model is misspecified by a factor α (the log-odds ratio of death between censored and uncensored patients at time t), then:

$$\text{Bias}(\alpha) \approx -\alpha\, \mathbb{E}\Big[\Lambda_0(C_i \mid \mathbf{x}_i)\, T_i\, S_{\text{full}}(\boldsymbol{\beta})\Big] \tag{4}$$

The bias is driven by α—the severity of the MAR violation—and the covariance between T_i and cumulative censoring hazard Λ_0. Because α cannot be identified from observed data alone, standard estimation fails, motivating the sensitivity analysis and frailty model. Our simulation demonstrates that the frailty approach faces identifiability challenges; the primary model (Model A) is given below alongside the frailty extension (Model B).

3.4 Logistic Hazard Model

The person-period format recasts discrete-time survival as a first-order Markov chain on $\{0, 1\}$ where 0 denotes "alive" and 1 denotes "dead" (absorbing state). The discrete-time hazard is the one-step transition probability, modeled as a Bernoulli GLMM with a logistic link.

Standard Model (Model A)

$$y_{i,t} \mid h_{i,t} \sim \text{Bernoulli}(h_{i,t}), \quad t = 1, \ldots, K_i,\ i = 1, \ldots, N, \tag{5}$$

$$\text{logit}(h_{i,t}) = \alpha_t + \mathbf{x}_i^\top \boldsymbol{\beta} \tag{6}$$

with priors:

$$\alpha_1 \sim \mathcal{N}(0, \sigma_0^2), \qquad \text{(anchor)} \tag{7}$$

$$\alpha_t \mid \alpha_{t-1} \sim \mathcal{N}(\alpha_{t-1}, \sigma_\alpha^2), \qquad t = 2, \ldots, K, \tag{8}$$

$$\sigma_\alpha \sim \text{Student-}t^+(\text{df}{=}3, 0, 1), \tag{9}$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}_P). \tag{10}$$

The first-order random walk prior on $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)^\top$ encodes the temporal correlation and the belief that neighboring intervals have similar baseline log-odds of death. As the risk set shrinks at later intervals, this prior borrows strength from adjacent time points, regularizing α_t where data is sparse. The half Student-t prior on σ_α is weakly informative and avoids the boundary issues of conjugate Inverse-Gamma priors with small shape parameters (Gelman, 2006). We set $\sigma_0^2 = \tau^2 = 10$ as weakly informative defaults.

Frailty Extension (Model B) To capture unobserved patient-level heterogeneity, we extend Model A with a correlation random effect b_i (frailty) for each patient i to account for the *within*-patient correlation across intervals:

$$\text{logit}(h_{i,t}) = \alpha_t + \mathbf{x}_i^\top \boldsymbol{\beta} + b_i \tag{11}$$

with $b_i \sim \mathcal{N}(0, \sigma_b^2)$ and $\sigma_b \sim \text{Exp}(1)$, called the penalized complexity (PC) prior, which shrinks towards no heterogeneity $b_i = 0$ (Simpson et al., 2017). By conditioning on b_i, the censoring assumption relaxes to $T_i \perp C_i \mid \mathbf{x}_i, b_i$—a strictly weaker condition than Assumption 3.2 (Daniels and Hogan, 2008). An important caveat is that addition of b_i does not guarantee $T_i \perp C_i \mid \mathbf{x}_i, b_i$ holds, as the frailty is a misspecified proxy for the true unobserved confounders that drive both censoring and failure. If b_i captures only partial heterogeneity, residual informative censoring bias remains. The practical downside is that N random effects are weakly identified from sparse binary panel data. The preliminary simulation confirms ESS that $\sigma_b \approx 80$–150.

3.5 Sensitivity Analysis via Exponential Tilting

The sensitivity analysis and frailty model address complementary aspects of informative censoring. The random effect b_i captures some heterogeneity *parametrically*, assuming the Gaussian prior (11) and the weaker conditional independence. The sensitivity analysis,

by contrast, *nonparametrically* aims to index the departures from independent censoring without specification of a particular structure.

Following Scharfstein et al. (1999) and the Bayesian treatment from Daniels and Hogan (2008), define γ as the log-odds ratio between patients censored at t and those remaining under observation:

$$\gamma = \log\left(\frac{\Pr(T_i = t \mid T_i \geq t, C_i = t, \mathbf{x}_i)}{\Pr(T_i = t \mid T_i \geq t, C_i > t, \mathbf{x}_i)}\right) \tag{12}$$

Here, $\gamma = 0$ corresponds to MAR, $\gamma < 0$ means censored patients have lower mortality risk (discharge reflects improvement), and $\gamma > 0$ means censored patients have higher mortality risk (discharge to hospice) (Oganisian, 2026). γ is equivalent to the selection bias parameter α in (4). The tilted hazard adjusts censored observations at their terminal interval:

$$\text{logit}(h_{i,t}) + \gamma \mathbb{I}(C_i = t, \delta_i = 0), \qquad \text{where } \delta_i = \mathbb{I}(T_i \leq C_i) \tag{13}$$

This addition to γ is an offset to the linear predictor. This multiplies by $\exp(\gamma)$ the odds of death for censored patients at their terminal interval.

MNAR model. Rather than fixing γ at a point-mass, we place a prior $\gamma \sim \mathcal{N}(0, \sigma_\gamma)$ and jointly estimate γ alongside $\boldsymbol{\beta}$. Since γ is also not identified from the data, the posterior $f(\gamma \mid \mathcal{D}^o)$ is heavily influenced by the prior. This is by design where the prior encodes substantive belief about plausible MNAR departures, and the posterior reflects the interaction between these beliefs and observed data likelihood. We assess prior sensitivity by fitting under different scales of $\sigma_\gamma \in \{0.5, 1.5, 2.0\}$ where the smaller σ_γ concentrates the prior more tightly around MAR while the larger σ_γ allows for more extreme departures. If the posterior for $\boldsymbol{\beta}$ is stable across prior scales, then clinical conclusions are robust to a wide range of MNAR assumptions.

4. Computation

4.1 Joint Posterior

The person-period likelihood combind with priors in Section 3.4 yields the joint posterior distribution for Model A:

$$f(\boldsymbol{\alpha}, \boldsymbol{\beta}, \sigma_\alpha \mid \mathcal{D}^o) \propto \underbrace{\prod_{i=1}^{N}\prod_{t=1}^{K_i} h_{i,t}^{y_{i,t}}(1-h_{i,t})^{1-y_{i,t}}}_{\text{Person-period likelihood}} \underbrace{\mathcal{N}(\boldsymbol{\alpha}; \mathbf{0}, \sigma_0^2)\prod_{t=2}^{K}\mathcal{N}(\alpha_t; \alpha_{t-1}, \sigma_\alpha^2)\, t_3^+(\sigma_\alpha)}_{\text{RW prior}} \mathcal{N}(\boldsymbol{\beta}; \mathbf{0}, \tau^2 \mathbf{I}_P) \tag{14}$$

where $h_{i,t} = \text{logit}^{-1}(\alpha_t + \mathbf{x}_i^\top \boldsymbol{\beta})$ and the likelihood factorizes over $R = \sum_i K_i$ person-period observations as independent Bernoulli trials. Model B extends this with N frailty terms and their hyperprior:

$$f(\boldsymbol{\alpha}, \boldsymbol{\beta}, \sigma_\alpha \mid \mathcal{D}^o) \propto \prod_{i,t} h_{i,t}^{y_{i,t}}(1-h_{i,t})^{1-y_{i,t}} f(\boldsymbol{\alpha} \mid \sigma_\alpha) f(\sigma_\alpha) f(\boldsymbol{\beta}) \prod_{i=1}^{N}\mathcal{N}(b_i; 0, \sigma_b^2)\text{Exp}(\sigma_b; 1) \tag{15}$$

where $h_{i,t} = \text{logit}^{-1}(\alpha_t + \mathbf{x}_i^\top \boldsymbol{\beta} + b_i)$. For the MNAR model, the tilted likelihood replaces the linear predictor with $\alpha_t + \mathbf{x}_i^\top \boldsymbol{\beta} + \gamma \mathbb{I}(C_i = t, \delta_i = 0)$ and appends the prior $\gamma \sim \mathcal{N}(0, \sigma_\gamma)$ giving the joint posterior over $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma)$.

4.2 Sampling Strategy

We initially developed a Pólya-Gamma (PG) Gibbs sampler (Polson et al., 2013) for Model A, which exploited the data-augmentation identity used in Polson et al. (2013) to obtain full conditional Gaussians for both $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, enabling efficient block Gibbs updates. For Model A this scheme converges efficiently. However, for Model B the centered parameterization exhibited Neal's funnel geometry (Neal, 2003), meaning when σ_b^2 is small the frailties b_i mix well, but for larger values the strong posterior correlation between σ_b and $\{b_i\}$ caused the sampler to stall, yielding very low ESS. This motivated the switch to Hamiltonian Monte Carlo (HMC) via Stan (Stan Development Team, 2024b).

4.3 HMC via Stan

We implement both models in Stan using the No-U-Turn Sampler (NUTS; Hoffman and Gelman, 2014), a self-tuning variant of HMC that adapts trajectory length during warmup via a recursive doubling procedure. Stan's non-centered parameterization ($b_i^{\text{raw}} \sim \mathcal{N}(0, 1)$) for Model B eliminates the funnel geometry. Combined with the PC prior on σ_b, this yields substantially better mixing. For both models, we run 4 chains with 500 warmup and 2,000 sampling iterations, yielding 8,000 posterior draws in CmdStanR (Stan Development Team, 2024a). Additionally, to avoid memory overflow from serializing 510K pointwise log-likelihoods per iteration, we fit "slim" models without the `generated_quantities` block and computed $\log p(y_r \mid \theta^{(s)})$ post-hoc in R.

For the MNAR sensitivity analysis, the joint posterior over $(\boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma)$ with $R > 500,000$ person-period rows makes full HMC prohibitively costly across three prior specifications. Instead, we use Stan's Automatic Differentiation Variational Inference (ADVI, full-rank) (Stan Development Team, 2024a), which approximates the posterior $p(\theta \mid \mathbf{y})$ with a multivariate Gaussian $q_\phi(\theta)$ by maximizing the evidence lower bound $\text{ELBO}(\phi) = \mathbb{E}_{q_\phi}[\log p(\theta, \mathbf{y}) - \log q_\phi(\theta)]$ via stochastic gradient ascent. While ADVI may underestimate posterior uncertainty in the tails, it suffices for recovering the posterior location and quantifying how $\hat{\boldsymbol{\beta}}$ shifts under MNAR, at orders of magnitude lower cost. The HMC results for Model A at $\gamma = 0$ serve as the gold-standard benchmark. The full code is available at `https://github.com/adamkurth/bayes-mimic-survival`.

5. Data Analysis: Application to MIMIC-III

5.1 Summary Statistics

The analytic cohort described in Section 2.1 comprises $N = 57,694$ first ICU admissions expanding to $R = 509,944$ person-period rows under daily discretization ($K = 30$). Table 3 summarizes covariate prevalence stratified by outcome. Key contrasts are: patients who died were older (mean standardized age 0.58 vs. 0.08), more frequently admitted emergently (93% vs. 82%), and had higher rates of sepsis (30% vs. 16%), respiratory failure (51% vs. 28%), and mechanical ventilation (58% vs. 25%). These disparities motivate the research

question posed in Section 1: whether these apparent risk factors remain significant once informative censoring is accounted for.

5.2 MCMC Convergence

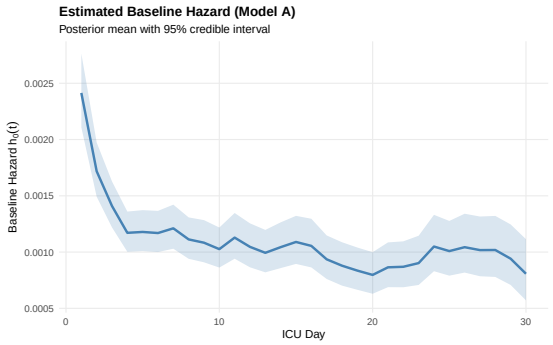
Table 2 reports full convergence diagnostics alongside posterior estimates. Model A converged cleanly with no divergent transitions across all 4 chains, all $\hat{R} < 1.01$, and minimum bulk ESS of 878 across all β_j. The random-walk intercepts α_t showed bulk ESS $> 1{,}500$ and tail ESS $> 1{,}200$, indicating adequate exploration of the posterior. The E-BFMI statistic exceeded 0.3 for all chains, confirming efficient energy transitions. Model B also produced no divergent transitions, but exhibited the identifiability challenges predicted by our simulation. The frailty standard deviation σ_b had ESS of only 97 with $\hat{R} = 1.03$, and E-BFMI flagged heavy tails in all chains. The fixed effects $\boldsymbol{\beta}$ under Model B had adequate ESS (> 500), though uniformly lower than under Model A. These diagnostics confirm that the $N = 57{,}694$ patient-level random effects are only weakly identified from sparse binary data, consistent with the simulation findings and reinforcing the motivation for the sensitivity analysis approach.

5.3 Results

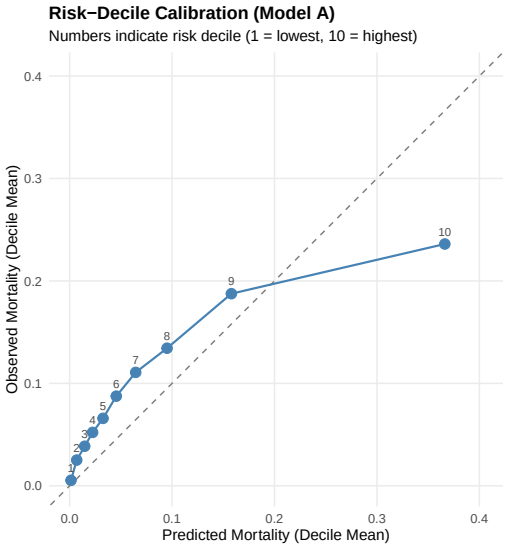
Model A (Standard Logistic Hazard). The estimated baseline hazard (Figure 2a) peaks on ICU day 1, declining steeply through day 5 and stabilizing near 0.001 thereafter, with widening credible intervals as the risk set depletes. This temporal profile is consistent with the acute-on-chronic risk period typical of ICU admissions where the highest mortality risk occurs immediately, with survivors exhibiting progressively better prognosis.

Figure 2b shows a risk-decile calibration plot with consistent over-prediction in deciles 1–9, consistent with the bias formula in (3): selective discharge of healthier patients inflates the predicted hazard for those remaining. Figure 3 overlays 200 posterior survival draws per model. The MAR cloud (grey) closely tracks the Kaplan–Meier estimate through day 10, while the three MNAR draw-clouds shift progressively upward, foreshadowing the sensitivity results.

Table 2 reports all posterior covariate effects (log-odds for daily mortality). The strongest risk factors were mechanical ventilation ($\hat{\beta} = 0.96$, 95% CrI [0.91, 1.02]), emergency admission (0.90, [0.81, 1.00]), standardized age (0.76, [0.72, 0.80]), ICD-coded malignancy (0.66, [0.61, 0.72]), GCS impairment (0.61, [0.55, 0.68]), and sepsis (0.50, [0.44, 0.56]). Pneumonia (-0.35) and positive blood cultures (-0.29) showed protective effects, likely reflecting that these diagnoses prompt early aggressive treatment with favorable short-term prognosis. Sex, COPD, vasopressor use, and dialysis showed credible intervals overlapping or near zero, suggesting minimal independent effects on daily mortality.



(a) Estimated baseline hazard $h_0(t) = \text{logit}^{-1}(\alpha_t)$ under Model A with 95% credible interval. The hazard peaks on day 1 and declines through the first week. Widening bands after day 20 reflect the shrinking risk set.



(b) Risk-decile calibration for Model A. Points above the diagonal indicate over-prediction, consistent with (3): selective discharge inflates the predicted hazard for remaining patients.

Figure 2: Model A diagnostics: **(a)** estimated baseline hazard over time and **(b)** risk-decile calibration.

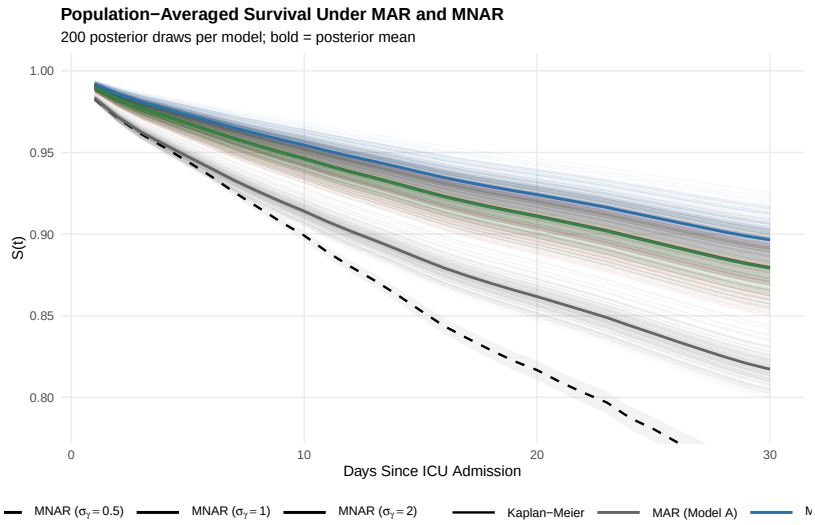


Figure 3: Population-averaged survival $S(t)$ under MAR and MNAR assumptions. Transparent curves show 200 posterior draws per model; bold lines show posterior means. The MAR draws (grey) and Kaplan–Meier (dashed black) closely track through day 10. Colored draw-clouds show MNAR-adjusted survival under $\sigma_\gamma \in \{0.5, 1.0, 2.0\}$. Wider priors produce both larger shifts and wider posterior spread.

Model B (Frailty as Comparator). The posterior mean $\hat{\sigma}_b = 1.38$ (95% CrI [1.17, 1.61]) indicates substantial unobserved heterogeneity. The 95th-percentile patient faces an odds of the event approximately 94 times higher ($\exp(3.29 \times 1.38) \approx 93.7$) than the 5th-percentile patient in any given day, holding covariates constant. This is clinically plausible given unmeasured factors such as pre-morbid functional status, goals-of-care decisions, and organ dysfunction severity not captured by ICD codes.

Including frailty uniformly amplified all covariate effects (Table 2), with largest shifts for ventilation (+0.28), sepsis (+0.20), malignancy (+0.19), and emergency admission (+0.17). This is classical attenuation bias for binary outcome where omitting a random effect biases coefficients toward zero (Daniels and Hogan, 2008). Model A's estimates therefore represent conservative lower bounds on the true covariate–hazard associations. However, the low ESS for σ_b (Section 5.2) means the frailty variance is only weakly identified, reinforcing the motivation for the sensitivity analysis over exclusive reliance on the frailty model.

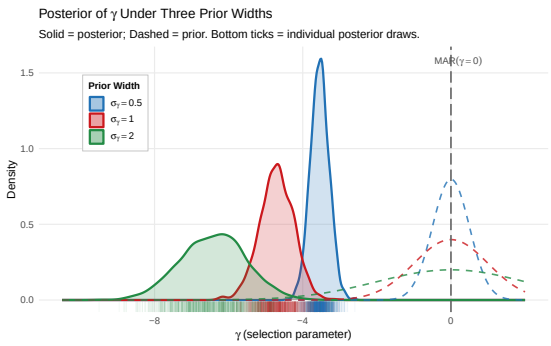
MNAR Sensitivity Analysis. Table 1 and Figure 4a present the posterior of γ under three prior scales. The posterior is decisively negative across all specifications: $\hat{\gamma} = -3.54$ (95% CrI $[-4.04, -3.06]$) for $\sigma_\gamma = 0.5$, $\hat{\gamma} = -4.71$ (CrI $[-5.64, -3.83]$) for $\sigma_\gamma = 1.0$, and $\hat{\gamma} = -6.42$ (CrI $[-8.13, -4.74]$) for $\sigma_\gamma = 2.0$. Because none of these credible intervals include zero, there is strong evidence that censoring is informative ($\gamma < 0$), supporting the hypothesis that discharged patients have systematically lower mortality risk than those remaining. The posterior mean shifts further from zero as the prior widens, illustrating that while the data provides a clear signal regarding the sign of the bias, it lacks the information to bound its magnitude and the posterior expands to reflect the scale of the chosen prior.

Under $\sigma_\gamma = 1.0$, the most affected covariate is emergency admission, whose effect attenuates from $\hat{\beta} = 0.902$ under MAR to 0.539 under MNAR ($\Delta\beta = -0.363$). Figure 4b isolates this covariate across prior widths, showing that the MAR posterior concentrates near 0.90 while the MNAR posteriors shift progressively leftward as σ_γ increases, reaching 0.56 under $\sigma_\gamma = 2.0$. Crucially, under this most permissive prior, the entire posterior distribution remains above zero, indicating that emergency admission remains a significant risk factor under all plausible departures from MAR.

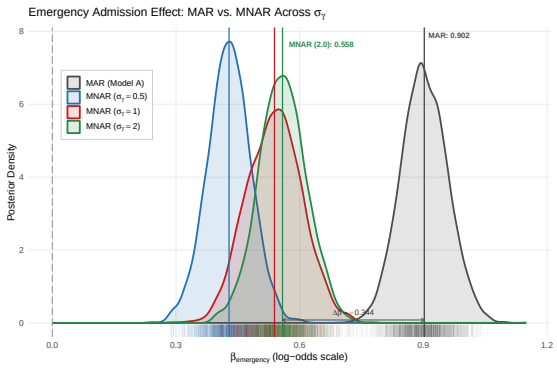
Other notable shifts include mechanical ventilation $\Delta\beta = -0.103$, GCS impairment -0.091, and malignancy -0.066. The pattern is: covariates associated with both higher mortality *and* longer ICU stays are most affected by the MNAR adjustment, because the selection parameter re-weighs the contribution of discharged patients whose unobserved outcomes would have lowered the apparent effect. Covariates near zero under MAR (vasopressor, COPD, dialysis) show negligible MNAR shifts ($|\Delta\beta| < 0.03$), indicating clinically ambiguous effects remain ambiguous regardless of the censoring assumption. The stability of the remaining 14 covariates across σ_γ values confirms that the core conclusions of the standard analysis are robust to a wide range of MNAR departures.

Conclusions. Returning to the central question posed in Section 1—*how robust are Bayesian posterior inferences about ICU mortality risk factors to departures from independent censoring?*—the answer is largely affirmative. All three analytical approaches (standard MAR, frailty, MNAR sensitivity) agree on the direction and rank ordering of covariate effects: mechanical ventilation, emergency admission, age, malignancy, GCS impairment, and sepsis are the dominant risk factors for 30-day ICU mortality. The frailty model and

MNAR analysis both indicate that Model A's MAR estimates are conservative, with true effects likely larger, but no covariate changes sign or loses significance under MNAR. Emergency admission is the most sensitive effect ($\Delta\beta = -0.36$ under $\sigma_\gamma = 1.0$), yet remains decisively positive. The calibration over-prediction (Figure 2b) is consistent with the bias direction predicted by (3), providing empirical corroboration of the theoretical framework. The MNAR posterior for γ is strongly negative across all prior scales, confirming that ICU discharge is indeed a positive prognostic signal.



(a) Posterior density of γ under three prior widths $\sigma_\gamma \in \{0.5, 1.0, 2.0\}$. Solid lines show posteriors; dashed lines show the $\mathcal{N}(0, \sigma_\gamma^2)$ priors. The vertical dashed line at $\gamma = 0$ marks MAR. All posteriors are decisively negative.



(b) Posterior density of $\beta_{\text{emergency}}$ under MAR (grey) and three MNAR specifications. The MAR estimate (0.902) attenuates to 0.558 under $\sigma_\gamma = 2$, yet remains decisively positive.

Figure 4: MNAR sensitivity analysis: **(a)** inferred selection parameter γ across prior widths, and **(b)** resulting attenuation of the emergency admission covariate effect.

Table 1: Posterior summary of γ under three prior widths σ_γ.

σ_γ	Mean	SD	95% CrI
0.5	-3.542	0.253	$[-4.041, -3.058]$
1.0	-4.705	0.458	$[-5.639, -3.829]$
2.0	-6.416	0.881	$[-8.134, -4.740]$

6. Discussion

This work makes three contributions to ICU survival analysis under informative censoring. First, we formalized the connection between independent censoring and MAR, deriving explicit bias formulas for hazard estimation and covariate effects. Second, we developed a sensitivity analysis using selection factorization that jointly estimates non-identifiable γ alongside $\boldsymbol{\beta}$ under proper priors, providing a principled assessment of robustness to MNAR departures. Third, through simulation and application to 57,694 MIMIC-III patients, we demonstrated that patient-level frailty is weakly identified, justifying the sensitivity analysis over direct frailty estimation.

Table 2: Posterior means, standard deviations (SD), effective sample sizes (ESS), and the frailty attenuation difference (Δ) for covariate effects. Both models exhibited excellent MCMC convergence (all $\hat{R} \leq 1.01$, zero divergent transitions). Model B additionally estimates the frailty standard deviation (σ_b).

	Model A (Standard)			**Model B (Frailty)**			
Covariate	**Mean**	**SD**	**ESS**	**Mean**	**SD**	**ESS**	Δ
σ_b (Frailty SD)	—	—	—	1.378	0.133	97	—
age (std)	0.757	0.026	5894	0.921	0.042	211	+0.163
male	−0.092	0.028	5256	−0.119	0.037	4463	−0.027
emergency	0.902	0.056	878	1.074	0.072	381	+0.172
icd_sepsis	0.502	0.035	5462	0.704	0.055	283	+0.202
icd_hf	−0.099	0.032	5469	−0.131	0.042	5186	−0.031
icd_resp	0.202	0.035	4925	0.263	0.046	2284	+0.061
icd_aki	0.193	0.032	6031	0.245	0.042	1845	+0.051
gcs_impaired	0.614	0.039	3584	0.718	0.051	530	+0.104
vasopressor	−0.063	0.035	5472	−0.035	0.047	4713	+0.028
icd_pneumonia	−0.353	0.034	5950	−0.493	0.050	547	−0.140
icd_diabetes	−0.139	0.032	7363	−0.188	0.041	3297	−0.049
icd_liver	0.436	0.048	5946	0.515	0.065	2273	+0.079
icd_malignancy	0.661	0.034	5393	0.852	0.054	294	+0.190
icd_copd	−0.044	0.038	7041	−0.068	0.051	5386	−0.025
is_ventilated	0.964	0.036	4520	1.250	0.067	162	+0.286
received_dialysis	0.046	0.047	6644	0.044	0.062	5174	−0.002
positive_blood_cx	−0.286	0.038	7050	−0.406	0.053	761	−0.120

The MNAR results present a coherent narrative. The decisively negative $\hat{\gamma}$ confirms that ICU discharge is indeed a positive prognostic signal, where discharged patients have systematically lower mortality risk, consistent with clinical intuition that clinicians discharge patients who are improving. The frailty model reaches complementary conclusions—including b_i uniformly amplifies all covariate effects, consistent with Model A absorbing unobserved heterogeneity into the baseline hazard and attenuating fixed effects toward zero. The two approaches agree on direction, but the standard model produces conservative estimates. The sensitivity analysis quantifies *how much* each covariate shifts under explicit MNAR departures, identifying emergency admission as the most sensitive effect and confirming that 14 of 17 covariates are robust to a wide range of MNAR assumptions.

Limitations. A major limitation is that we include only static baseline covariates. Time-varying covariates (daily labs, vasopressor dosing) are likely informative for both mortality and discharge decisions, and their inclusion could reduce the degree of informative censoring. The sensitivity analysis assumes a constant γ across all censoring times and patient profiles; in practice, the selection mechanism may vary by time and patient characteristics. Extensions to covariate-dependent $\gamma(\mathbf{x}_i)$ are possible within this framework but increase the dimensionality of the sensitivity space. The MNAR model is fit using ADVI rather than full HMC, which may underestimate posterior uncertainty in tails.

Future work. Competing risks of death vs. discharge represent the most natural extension. In the discrete-time framework, this corresponds to a multinomial rather than

Bernoulli observation model at each interval with separate hazards for each competing event, and the exponential tilting framework extends naturally by introducing event-specific sensitivity parameters γ_k.

Table 3: Baseline characteristics and covariate definitions ($\mathbf{X}$) for the MIMIC-III analytic cohort within the 30-day observation window, stratified by outcome. *Discharged* denotes patients censored alive before the window closes (the clinically informative censoring mechanism); *Survived* denotes patients still alive at 30 days (administratively censored). Values are n (%) for categorical variables and mean ± SD or median [IQR] for continuous variables.

	Overall (N=57,694)	**Died** (n=5,443)	**Discharged** (n=49,660)	**Survived** (n=2,591)
Demographics & Admission				
Age, years (std)[a]	55.1 ± 27.3	69.8 ± 17.2	54.2 ± 27.3	40.2 ± 31.1
Male sex	32,248 (55.9%)	2,906 (53.4%)	27,869 (56.1%)	1,473 (56.9%)
Emergency admission	41,580 (72.1%)	5,081 (93.3%)	34,969 (70.4%)	1,530 (59.1%)
Follow-up, days	5.9 [3.5, 10.6]	4.5 [1.6, 9.8]	5.7 [3.6, 9.6]	42.0 [35.0, 57.3]
Comorbidities [ICD-9 Codes]				
Sepsis [995.91/92, 038.x][b]	6,320 (11.0%)	1,777 (32.6%)	3,878 (7.8%)	665 (25.7%)
Heart failure [428.x][b]	13,434 (23.3%)	1,840 (33.8%)	11,036 (22.2%)	558 (21.5%)
Acute kidney injury [584.x][b]	11,359 (19.7%)	2,335 (42.9%)	8,176 (16.5%)	848 (32.7%)
Respiratory failure [518.8x][b]	8,906 (15.4%)	2,460 (45.2%)	5,661 (11.4%)	785 (30.3%)
Pneumonia [480–486]	7,347 (12.7%)	1,332 (24.5%)	5,322 (10.7%)	693 (26.7%)
Diabetes [250.x]	13,900 (24.1%)	1,423 (26.1%)	12,078 (24.3%)	399 (15.4%)
Liver disease [571.x]	2,916 (5.1%)	538 (9.9%)	2,195 (4.4%)	183 (7.1%)
Malignancy [140–208]	6,730 (11.7%)	1,258 (23.1%)	5,184 (10.4%)	288 (11.1%)
COPD [490–492, 494, 496]	6,568 (11.4%)	870 (16.0%)	5,508 (11.1%)	190 (7.3%)
Clinical Indicators [Procedure Codes]				
GCS impaired (≤ 12)[b]	30,190 (52.3%)	4,495 (82.6%)	24,358 (49.0%)	1,337 (51.6%)
Mechanical ventilation [96.7x]	14,477 (25.1%)	3,492 (64.2%)	9,240 (18.6%)	1,745 (67.3%)
Vasopressor use[b]	6,836 (11.8%)	1,312 (24.1%)	5,134 (10.3%)	390 (15.1%)
Dialysis [39.95]	3,166 (5.5%)	596 (10.9%)	2,195 (4.4%)	375 (14.5%)
Positive blood culture	5,448 (9.4%)	1,049 (19.3%)	3,546 (7.1%)	853 (32.9%)

[a] Age was standardized to $\mu = 0, \sigma = 1$ for inclusion in the design matrix.

[b] Variables explicitly included in the simulation study.

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